

AI DRIVEN DESIGN VALIDATOR SOFTWARE

VENDOR INFORMATION, CLARIFICATION REQUIREMENTS AND MVP COMPLIANCE MATRIX

1. General Instructions

- a) Vendors shall fill all columns and provide detailed responses.
- b) Vendors shall clearly indicate the features:
 - Available in Current Product
 - Available with Customization
 - Requires New Development
 - Not Available
- c) Documentary evidence shall be attached wherever applicable.
- d) Screenshots of MVP demonstrations shall be submitted.
- e) Sample output files generated during MVP demonstration shall be submitted in(as applicable)
 - MS Word Format (.docx)
 - MS Excel Format (.xlsx)
 - PDF Format (.pdf)
- f) Vendors shall clearly indicate assumptions, limitations and dependencies.
- g) Supporting documents such as work completion certificates, client references, shall be attached.

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2. The following details to be provided

PART-A : VENDOR ORGANIZATION DETAILS

SI No	Particulars	Vendor Response
1.	Name of Company	
2.	Registered Office Address	
3.	Development Centre Location(s)	
4.	Year of Establishment	
5.	Company Type (MSME/Startup/Large Enterprise)	
6.	Registration Number	
7.	GST Number	
8.	Total AI/ML Engineers	
9.	Total Software Developers	
10.	Total Domain Experts	
11.	Total Employees Proposed for this Project	
12.	Number of Personnel Proposed to be Deployed at HSL	
13.	Number of Personnel Available for Remote Support	

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PART-B : EXPERIENCE IN SIMILAR AI PROJECTS AND DOMAIN EXPERTISE

Vendors shall provide details of Artificial Intelligence (AI), Generative AI (GenAI), Machine Learning (ML), Knowledge Management, Document Intelligence, Engineering Intelligence, Retrieval-Augmented Generation (RAG), and other related projects successfully executed by the organization.

Particular emphasis shall be placed on projects involving Engineering Design Assistance, Rules & Regulations Interpretation, Technical Document Intelligence, Knowledge Repositories, Technical Specification Generation, Engineering Drawing Intelligence, Technical Offer Evaluation, Inspection Report Analytics, Lessons Learnt Management and Decision Support Systems.

For MSMEs & large companies, submission of work experience certificate on similar AI project is mandatory. Additional weightage will be considered for projects executed for Government Organizations, Defence Establishments, Public Sector Undertakings (PSUs), Shipyards or other large enterprises handling complex technical documentation and knowledge management requirements.

Important Notes

1. Vendors shall furnish details only of successfully completed, accepted and deployed projects. Projects under development, implementation, pilot stage, proof-of-concept (PoC), MVP stage or currently under execution shall not be considered as implemented projects for technical evaluation.
2. Vendors shall submit documentary evidence for each claimed project such as Work Completion Certificates, Client Acceptance Certificates, Go-Live Certificates, Purchase Orders with Completion Confirmation or equivalent proof.
3. Vendors shall clearly indicate whether the referenced project was deployed in a Production Environment, Pilot Environment, or Testing Environment. Only Production Deployments shall be considered as fully implemented projects.
4. HSL reserves the right to seek additional documentary evidence, customer references, demonstrations, or clarifications to verify the claimed experience.
5. Any project for which adequate documentary evidence is not furnished may not be considered during technical evaluation.

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6. Vendors shall provide details of technologies used in each project, including LLMs, OCR engines, RAG frameworks, Vector Databases, AI/ML frameworks and deployment architecture.
7. Preference will be given to vendors possessing demonstrable experience in developing and deploying AI/GenAI solutions in engineering, manufacturing, shipbuilding, defence, government, PSU, and technical knowledge management domains.

Sl. No.	Particulars	Vendor Response
1.	Total Number of AI/GenAI Projects Executed	
2.	Total Number of AI/GenAI Projects Executed for Government Organizations/PSU	
3.	Total Number of AI/GenAI Projects Executed in Engineering/Manufacturing Domain	
4.	Total Number of Knowledge Management / RAG-Based Solutions Implemented	
5.	Total Number of Document Intelligence / OCR-Based Solutions Implemented	
6.	Total Number of Engineering Drawing Intelligence Solutions Implemented	
7.	Total Number of Chatbot / AI Assistant Solutions Implemented	
8.	Experience in On-Premise / Air-Gapped AI Deployments (Provide Details)	

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Details of Relevant Projects

Sl. No.	Client Name	Project Title	Brief Scope of Work	Domain (Govt/PSU/Defence/Industry)	Year of Completion	Current Status	Technologies Used (LLM/OCR/RAG etc.)	Work Completion Certificate Attached (Yes/No)
1								
2								
3								
4								
5								

Experience Relevant to AI Driven Design Validator

Sl. No.	Capability Area	Similar Project Executed (Yes/No)	Project Reference / Remarks
1	Rules & Regulations Chatbot / AI Assistant		
2	Knowledge Repository / RAG Solution		
3	SOTR / Specification Generation		
4	BOM Generation		

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Sl. No.	Capability Area	Similar Project Executed (Yes/No)	Project Reference / Remarks
5	Pre-Bid Query Generation		
6	Technical Offer Evaluation		
7	Binding Data Scrutiny		
8	Intelligent Document Conversion		
9	Inspection Report Analytics		
10	Lessons Learnt Management System		
11	Engineering Drawing Intelligence		
12	Design Review Assistant		
13	DWG/CAD Drawing Processing		
14	Enterprise Search & Knowledge Management		
15	On-Premise AI Deployment		
16	Air-Gapped Deployment		

Supporting Documents

Vendors shall submit the following documents as evidence:

1. Work Completion Certificates.

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2. Client Appreciation/Reference Letters (if available).
3. Client References for verification

PART-C : AI ARCHITECTURE DETAILS

SI No	Particulars	Vendor Response
1	LLM Used in MVP	
2	LLM Proposed for Final Deployment	
3	Open Source / Proprietary	
4	OCR Engine Used in MVP	
5	OCR Engine Proposed for Final Deployment	
6	Vector Database Proposed	
7	RAG Framework Used	
8	Proposed architecture and database technology for secure user management, access control, auditability, activity tracking, and application data management	
9	Hallucination Control Mechanism	
10	Source Traceability Mechanism	

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SI No	Particulars	Vendor Response
11	Internet Required (Yes/No)	
12	Fully On-Premise Deployment Possible (Yes/No)	
13	Air-Gapped Deployment Possible (Yes/No)	

PART-D : DEVELOPMENT AND DEPLOYMENT PLAN

SI No	Activity	Time Required
1	Requirement Study	
2	Data Collection and Preparation	
3	Knowledge Base Creation	
4	AI Model Training	
5	Application Development	
6	Integration Activities	
7	User Acceptance Testing	
8	Deployment	
9	User Training	

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SI No	Activity	Time Required
10	Total Project Duration	

PART-E: FUNCTIONALITIES COMPLIANCE MATRIX

S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
1.	Rules & Regulations Engineering Assistant	System shall be trained on IRS Rules and Regulations. It shall answer user queries, provide source references, lessons learnt based suggestions, support follow-up questions and maintain feedback mechanism for each response. : Use IRS main rules provided by HSL or from open source	System shall be trained on complete organizational knowledge base and act as an Engineering Expert Assistant using Rules, Regulations, Lessons Learnt, Manuals, Build Specifications, Historical Projects, Inspection Reports and Trial Reports.					
2.	BOM Generation from	The AI tool shall demonstrate the capability to generate a preliminary Bill of	The system shall function as an intelligent engineering assistant capable of generating complete Bills of Materials (BoMs) by					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
	RFP/Build specification	Materials (BoM) by understanding the Build Specification/RFP and applying engineering reasoning, historical project knowledge, engineering calculations, and design assumptions. The system shall be capable of: a) Reading and understanding the uploaded Build Specification/RFP. b) Identifying all systems and sub-systems specified in the document. c) Automatically grouping BoM	integrating Build Specifications, Principal Particulars, Historical Project Data, Engineering Calculations, Functional Design Logic, Rules & Regulations, Technical Manuals and Engineering Standards. The system shall be capable of: a) Automatically identifying all ship systems and sub-systems from the Build Specification. b) Generating complete system-wise and discipline-wise BoMs in the prescribed HSL format. c) Estimating quantities of equipment, materials, cables, cable accessories, piping items, fittings, supports, instruments, and installation materials where quantities are not explicitly specified.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		items discipline-wise (Electrical, Machinery, Hull, Outfit, Piping etc). d) Generating a preliminary system-wise and discipline-wise BoM in the prescribed HSL format. e) Extracting equipment descriptions, technical specifications, units, quantities (where available), and remarks from the specification. f) Identifying items whose quantities are not explicitly	d) Deriving preliminary functional layouts and engineering design logic using vessel Principal Particulars, General Arrangement (GA) drawings, compartment information, historical vessel designs, and engineering standards to support quantity estimation. e) Performing engineering calculations required for sizing, capacity determination, and quantity estimation of ship systems and equipment. f) Estimating quantities for all major ship systems, including Electrical, Machinery, Hull, Outfit, Piping, HVAC etc. g) Estimating cable quantities, cable trays supports and related installation materials					

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S.N o	Feature	HSL Expectation During MVP	HSL Expectation After Development	Curr ent Pro duct	Custom ization Require d	New Develo pment require d	Vendor complan	Remarks
		mentioned and indicating them for engineering estimation. g) Demonstrating engineering reasoning by estimating quantities for selected systems using engineering calculations, applicable rules and design assumptions. h) Demonstrating one or more engineering calculation-based use cases such as: Electrical Load Analysis for	using equipment locations, routing philosophy, historical project data and engineering design rules. h) Providing complete traceability for every estimated quantity, including engineering calculations, assumptions, historical references, applicable rules, and source documents. i) Allowing designers to review, modify, and approve AI-generated quantities before finalization of the BoM. j) Continuously improving estimation accuracy through newly added project data, engineering calculations, lessons learnt, and user feedback.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Generator Capacity Estimation - Engine/Equipment Capacity Estimation based on functional requirements. - Estimation of Public Address (PA) System Speakers using General Arrangement (GA) drawings, compartment type,	Illustrative Use Cases - Electrical Load Analysis and Generator Capacity Estimation. - Engine Capacity Estimation based on operational requirements. - Estimation of Public Address (PA) System Speakers based on GA drawings and compartment information. - Estimation of Lighting Fixtures based on compartment dimensions and illumination criteria. - Estimation of Fire Detection Devices based on compartment type and applicable standards. - Estimation of Cable Quantities and Cable Accessories based on equipment locations and routing philosophy.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		compartment dimensions, and applicable design criteria. - o Estimation of selected equipment /material quantities based on engineering calculations i) Clearly indicating the basis of estimation, assumptions, engineering calculations, historical	• Estimation of cable trays, Supports, and Installation Materials. • Generation of complete discipline-wise BoMs for Electrical, Machinery, Hull, Outfit, Piping, HVAC, and other ship systems. <i>Expected Outputs</i> • Complete System-wise BoM. • Complete Discipline-wise BoM. • Quantity Estimation Report. • Engineering Calculation & Assumption Report. • Design Basis Report. • Source Traceability Report. • Excel, Word, and PDF Outputs.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		references, and applicable rules used for every estimated quantity. j) Providing complete source traceability for all extracted and estimated data. k) Allowing export of the generated BoM in Excel, Word, and PDF formats. <i>Expected Outputs</i> • System-wise BoM. • Discipline-wise BoM. • Preliminary Quantity						

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Estimation Report. - Engineering Calculation & Assumption Report. - Source Traceability Report. - Excel, Word, and PDF Outputs.						
3.	SOTR (Specification of Technical Requirement / Tender Technical specification)Generation	The AI tool shall demonstrate the capability to generate a preliminary SOTR for a selected system in the prescribed HSL format by utilizing the uploaded Build Specification/RFP, applicable Rules & Regulations, and the quantities derived as	The system shall function as an intelligent engineering assistant capable of automatically generating complete SOTRs/Tender Technical Specifications by utilizing the Build Specification/RFP, AI-generated BoM, applicable Rules & Regulations, Technical Manuals, Historical Project Data, Lessons Learnt, Inspection Reports, Engineering Standards					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		part of the BoM Generation module. The system shall be capable of: (a) Reading and understanding the uploaded Build Specification/RF P. (b) Identifying the selected system and its functional requirements. (c) Utilizing the preliminary BoM generated for the selected system, including the estimated quantities of equipment and materials, while preparing the SOTR.	and other HSL knowledge repositories. The system shall be capable of: (a) Automatically generating complete system-wise SOTRs in the prescribed HSL format. (b) Utilizing the quantities derived through the BoM Generation module for every applicable equipment, subsystem, cable, piping items, fitting, support, and installation material while preparing the SOTR. (c) Automatically generating complete technical specifications, performance criteria, functional requirements, inspection and testing requirements, documentation requirements, acceptance					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		(d) Automatically incorporating technical requirements, performance requirements, functional requirements, inspection requirements, testing requirements, documentation requirements, acceptance criteria, and equipment/material quantities into the SOTR. (e) Incorporating applicable Rules & Regulations with appropriate references. (f) Ensuring consistency between the	criteria, mandatory vendor deliverables and compliance requirements for all equipment and systems included in the SOTR. (d) Ensuring complete technical consistency between the Build Specification, AI-generated BoM, applicable Rules & Regulations and the generated SOTR. (e) Identifying missing technical requirements and recommending additional clauses based on Rules & Regulations, Historical Project Data and Lessons Learnt. (f) Providing complete traceability for all technical requirements by indicating the applicable source documents, rules,					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Build Specification, BoM, and generated SOTR. (g) Exporting the generated SOTR in Word and PDF formats. <i>Expected Outputs</i> • System-wise SOTR. • Rule & Regulation Compliance Matrix. • Technical Requirement Traceability Matrix. • Source Traceability Report.	standards, manuals, and historical references. (g) Allowing designers to review, modify, and approve the AI-generated SOTR before finalization. (h) Continuously improving SOTR generation through newly added project data and user feedback. <i>Expected Outputs</i> • Complete System-wise SOTR/Tender Technical Specification. • Rule & Regulation Compliance Matrix. • Technical Requirement Traceability Matrix. • Source Traceability Report. • Word and PDF Outputs.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Word and PDF Outputs.						
4.	Pre-Bid Query Generation	Generate Pre-Bid Queries from RFP and identify ambiguities, inconsistencies, missing information and technical risks. (use Build specification of a Tug and sample format shared by HSL)	The AI tool shall be trained using historical Pre-Bid Queries and historical Project Data. The system shall be capable of: - Automatically analyzing RFPs to generate discipline-wise pre-bid queries. - Identifying ambiguities, inconsistencies, conflicting requirements, missing information, incomplete specifications, and potential technical risks. - Learning from historical HSL pre-bid queries, customer clarifications, and lessons learnt to improve the quality and relevance of generated queries.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			d) Generating queries related to design, procurement, manufacturing, installation, testing, commissioning, integration, documentation, spares, training, and compliance requirements. e) Categorizing queries discipline-wise (Electrical, Machinery, Hull, Piping, Outfit etc.). f) Providing references to relevant clauses, rules, standards, manuals, lessons learnt, and historical project records used for generating the queries. g) Identifying areas that may result in cost overruns, schedule delays, design changes, contractual issues, procurement challenges, or operational					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			risks if clarification is not obtained. h) Generating a structured and traceable query register with supporting justification. i) Allowing users to review, modify, add, or regenerate queries based on prompts. j) Exporting outputs in Excel, Word, and PDF formats. The system shall function as an intelligent Pre-Bid Review Assistant, helping engineers identify technical gaps, risks, and clarification requirements at an early stage by leveraging organizational knowledge, historical experience and applicable standards.					
5.	Technical Offer Evaluation	Upload Technical Offer and TTS/SOTR. Generate Technical Compliance Matrix.	Comparing Vendor Technical Offers against TTS/SOTR clause-by-clause.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Identify deviations, exclusions and missing information. Suggest technical queries. (use SOTR of HDCS and Technical offer of HDCS)	☐ Automatically generating Technical Compliance Matrices. ☐ Identifying: • Complied Clauses • Partially Complied Clauses • Non-Complied Clauses • Clauses Not Addressed by Vendor • Deviations and Exceptions • Missing Information ☐ Generating technical clarification points to be raised with vendors. ☐ Performing risk assessment of deviations and non-compliances. ☐ Comparing multiple vendor offers and generating comparative evaluation reports. ☐ Providing AI-based recommendations with justification and source references. ☐ Learning from historical evaluation reports and vendor performance records to improve					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			the quality of evaluation and recommendations. ☐ Exporting outputs in Excel, Word, and PDF formats. ☐ Providing complete traceability by indicating the source document, clause number and basis for each observation or recommendation.					
6.	Binding Data Gap Analysis & Information Extraction Assistant.	a. Compare Binding Data against POTS and identify missing documents and deviations b. Extraction of required information from the Binding data based on user prompt and option to export into required format(Use Binding data and POTS of MB/SRE ,output format for extraction of equipment shared by HSL)	The AI tool shall be capable of: a) Comparing Vendor Binding Data against POTS requirements & Functional drawing and identifying missing documents, missing information, deviations, non-compliances and incomplete submissions. b) Automatically generating a Binding Data Compliance Matrix indicating required documents, submitted documents, compliance status and remarks.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			c) Generating clarification points to be raised with vendors for missing or non-compliant submissions. d) Extracting information from Binding Data based on user prompts and generating structured outputs in user-defined formats. e) Understanding and extracting information from manuals, datasheets, drawings, certificates, calculations, and test reports. f) Generating customized reports such as equipment lists, cable schedules, instrument lists, spare parts lists, technical parameter sheets, document registers, and certification matrices. g) Providing traceability by indicating the source					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			document and reference from which information has been extracted. h) Learning from historical Binding Data reviews and lessons learnt to improve scrutiny and information extraction accuracy. i) Exporting outputs in Excel, Word, and PDF formats. The system shall function as an intelligent Binding Data Review Assistant, significantly reducing manual effort in scrutiny, compliance verification and extraction of engineering information from vendor documentation.					
7.	Intelligent Document Converter	The system shall be capable of: a) Accepting single or multiple document uploads,	The system shall be capable of: (a) Processing user-uploaded or user-selected documents from the workspace.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		including PDF, scanned documents, images, and office documents. b) Allowing users to provide natural language prompts specifying the required output. c) Understanding document content using AI-based document intelligence and not merely performing simple OCR conversion. d) Extracting structured information from specifications, technical manuals,	(b) Extracting information from documents based on natural language prompts. (c) Generating customized outputs with user-defined columns and formats in Excel, Word and PDF. (d) Consolidating information from multiple selected/uploaded documents into a single output. (e) Comparison of documents feature to be available as per user prompt (f) Supporting batch processing of multiple documents and generation of structured reports. The system shall function as an intelligent document understanding and information extraction assistant, significantly reducing manual effort in					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		datasheets, reports, drawings, and tabular documents. e) Generating customized outputs with user-defined columns and formats in Excel, Word, and PDF. f) Comparison of documents feature to be available as per user prompt g) Supporting multi-document processing and consolidation of information from multiple uploaded files. (use images provided by HSL)	preparing structured engineering documents and reports.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
8.	Inspection Report Analytics	The AI tool shall be capable of reading and analyzing uploaded inspection reports and classifying observations/remarks into categories such as Design Related, Material Related, Workmanship Related, Documentation Related, and Testing & Commissioning Related. The system shall be capable of: (a) Extracting and categorizing observations,	The AI tool shall be trained using Inspection Reports, Trial Reports, Historical Project Data, and related project records. The system shall be capable of: (a) Reading and analyzing inspection reports and automatically classifying observations/remarks into categories such as Design Related, Material Related, Workmanship Related, Documentation Related, and Testing & Commissioning Related etc (b) Identifying the associated project, system, discipline, inspection agency, and observation status. (c) Generating system-wise, discipline-wise, project-wise, and agency-wise analytics and reports.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		remarks, and deficiencies from inspection reports. (b) Generating system-wise, discipline-wise, project-wise, and agency-wise analytics. (c) Answering user queries based on inspection report data. Examples: - How many Workmanship Related remarks exist in Project XXX? - How many UNSAT remarks were raised by WATT in Project XXX? - List all Material Related remarks for MB/SRE System.	(d) Answering user queries based on inspection report data. Examples: - How many Workmanship Related remarks exist in Project XXX? - How many UNSAT remarks were raised by WATT in Project XXX? - List all open remarks related to MB/SRE System. 5. Allowing authorized users to update observation status as Open, Under Action, Closed, or Deferred, with complete audit trail. 6. Providing configurable dashboards and analytics based on project, system, discipline, agency, status, category, and date range. 7. Providing source traceability and references for all					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		(d) Allowing authorized users to update the status of observations as Open, Under Action, Closed, or Deferred, with complete audit trail. (e) Providing configurable dashboards and analytics based on project, system, discipline, agency, status, and date range. (f) Providing source references for generated observations and reports. (g) Exporting outputs in Excel,	observations and analytics. 8. Learning from historical inspection reports and user feedback to improve classification accuracy. 9. Exporting outputs in Excel, Word and PDF formats. Expected Outputs • Categorized Inspection Remark Register. • Open/Closed Remark Status Register. • System-wise, Discipline-wise, Project-wise, and Agency-wise Analytics Reports. • Dashboard Analytics with Closure Status. • User Query Responses with Source References. • Excel, Word, and PDF Reports. (The format and the details to be included in this modules will be					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Word, and PDF formats.	intimated during development stage by the User Department) The system shall function as an intelligent Inspection Report Analytics and Remark Management Assistant to support remark tracking, closure monitoring, trend analysis and project review activities. .					
9.	Lessons Learnt Repository	The AI tool shall demonstrate the capability to create and query a Lessons Learnt Repository using historical project documents, inspection reports, technical correspondence, vendor issues and other relevant records provided for the MVP.	The AI tool shall function as an organizational knowledge repository and intelligent engineering assistant by leveraging Lessons Learnt from historical projects, inspection reports, design reviews, technical correspondence, vendor issues, operational feedback, technical deviations and other project records available within HSL. The system shall be capable of: a) Creating and maintaining a centralized, searchable Lessons Learnt					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			Repository across all projects and disciplines. b) Generating project-wise, system-wise, discipline-wise, equipment-wise, and vendor-wise lessons learnt based on user prompt c) Providing proactive design recommendations, precautions, and best practices based on historical project experience. d) Correlating lessons learnt with applicable Rules & Regulations, Technical Manuals, SOTRs, Inspection Reports etc e) Answering user queries and providing relevant lessons learnt Examples: What lessons learnt should be considered while designing the MB/SRE System?					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			- What recurring issues were observed in HDCS systems across previous projects? - What lessons learnt are applicable to communication systems of X project? - What vendor-related issues were encountered in previous projects for a particular equipment? 6. Providing recommendations to prevent recurrence of historical design, material, workmanship, documentation, testing and integration issues. 7. Identifying recurring trends, common failures, and frequently observed deficiencies across projects. 8. Based on data from SAP(if integration is not possible information in excel will be					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			loaded periodically) vendor delivery issues to be identified whether supplying on time or delay is there and recommendations are to be made accordingly. 9. Continuously enriching the repository through newly added project data and approved user feedback. 10. Providing complete source traceability and references for all generated lessons learnt and recommendations. 11. Exporting outputs in Excel, Word, and PDF formats. Expected Outputs • Project-wise Lessons Learnt Register. • System-wise Lessons Learnt Report.					

AI DRIVEN DESIGN VALIDATOR SOFTWARE

VENDOR INFORMATION, CLARIFICATION REQUIREMENTS AND MVP COMPLIANCE MATRIX

S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			• Discipline-wise Lessons Learnt Report. • Equipment-wise Lessons Learnt Report. • Vendor-wise Lessons Learnt Report. • Design Recommendations and Best Practices Report. • Trend Analysis and Recurring Issues Report. • User Query Responses with Source References. • Excel, Word, and PDF Reports. The system shall function as an intelligent organizational knowledge assistant, enabling engineers to leverage historical experience, avoid repeat issues, improve design quality, and support informed decision-making throughout the project lifecycle.					

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10	Engineering Drawing Intelligence	The AI tool shall demonstrate the capability to read and understand engineering drawings in PDF and DWG formats and perform basic intelligence-based operations. The system shall be capable of: (a) Accepting engineering drawings in PDF and DWG formats from workspace or user upload. (b) Extracting and interpreting key information from drawings such as equipment, tags, connections, and annotations. (c) Understanding basic drawing	The system shall function as an intelligent Engineering Drawing Validator and Information Extraction Assistant by leveraging HSL engineering drawings, Rules & Regulations, Build Specifications, Binding data, SOTRs/TTS, POTS, Technical Manuals, Lessons Learnt, Inspection Reports, and historical project data. (complete knowledge base) The system shall be capable of: (a) Reading, understanding, and processing engineering drawings in PDF and DWG formats. (b) Validating drawings against applicable Rules & Regulations, Build Specifications, Binding data of corresponding equipment /system, Lessons Learnt, and approved engineering practices.					

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VENDOR INFORMATION, CLARIFICATION REQUIREMENTS AND MVP COMPLIANCE MATRIX

S.N o	Feature	HSL Expectation During MVP	HSL Expectation After Development	Curr ent Pro duct	Custom ization Require d	New Develo pment require d	Vendor complan	Remarks
		structure (schematics, layouts, interconnection diagrams where applicable). (d) Validating drawings against applicable: ○ Rules & Regulations (use IRS rules for testing) ○ Build Specifications (use Tug document given by HSL) ○ Lessons Learnt (limited dataset for MVP)	(c) Identifying non-compliances, design inconsistencies, missing information, interface mismatches and potential design risks, along with applicable references. (d) Providing intelligent design recommendations, review comments, and corrective actions to assist designers during design and review stages. (e) Extracting engineering information from drawings based on user prompts and generating structured outputs in user-defined formats. Example User Prompts: • Generate Cable Schedule for MB/SRE System. • Generate Equipment List from the uploaded drawing. • Extract Junction Box Details.					

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VENDOR INFORMATION, CLARIFICATION REQUIREMENTS AND MVP COMPLIANCE MATRIX

S.N o	Feature	HSL Expectation During MVP	HSL Expectation After Development	Curr ent Pro duct	Custom ization Require d	New Develo pment require d	Vendor complan	Remarks
		○ Binding data of equipment (use the document given by HSL) (e) Identifying missing or inconsistent information in drawings. (f) Allowing user-driven extraction of information based on prompts. Example User Prompt: • “Generate Cable Schedule for MB/SRE System” • Make compartment list with area of each compartment	• Generate Interconnection Schedule. • List all equipment associated with a particular system. (f) Generating user-defined outputs such as Cable Schedules, Equipment Lists, Interconnection Lists, Cable Routing Data, Junction Box Schedules, and other engineering reports. (g) Consolidating and correlating information from multiple related drawings for system-level analysis. (h) Providing source traceability with drawing references and applicable rule references. (i) Learning from newly added drawings, user feedback, and project data to continuously improve					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		deck wise from GA of Project X Expected Output: • Cable Tag • From Equipment & Location • To Equipment & Location • Cable Size • Cable Type (g) Providing source traceability (drawing name, sheet reference). (h) Exporting output in Excel / Word formats.	validation and extraction accuracy. (j) Exporting outputs in Excel, Word, and PDF formats. The system shall act as an intelligent engineering assistant for drawing validation, design review, compliance verification, and extraction of structured engineering data from engineering drawings.					
11	Design Review Assistant for Checklists & Risk Analysis	The AI tool shall demonstrate the capability to generate design review checklists and identify potential design risks based on the uploaded documents and	The system shall function as an intelligent Design Review Assistant by leveraging Rules & Regulations, Build Specifications, SOTRs/TTS, Technical Manuals, Lessons Learnt, Inspection Reports, Historical Project Data, and					

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VENDOR INFORMATION, CLARIFICATION REQUIREMENTS AND MVP COMPLIANCE MATRIX

S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		knowledge base used for MVP demonstration. The system shall be capable of: a) Generating system-specific design review checklists based on user prompts. b) Identifying key design risks, missing checks, and critical review points. c) Providing references to applicable Rules & Regulations and supporting documents. d) Answering user queries related to design review requirements. Example User Prompts:	Engineering Drawings (complete knowledge base). The system shall be capable of: a) Generating project-wise, system-wise, and discipline-wise design review checklists based on user prompt. b) Identifying potential design risks, compliance gaps, interface issues, and critical review points. c) Providing risk mitigation recommendations and best practices based on historical project experience. d) Correlating checklist items with applicable Rules & Regulations and Lessons Learnt.etc e) Supporting design reviews for drawings, specifications, technical documents, and engineering deliverables.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		- Generate Design Review Checklist for MB/SRE System. - What are the key risks to be considered while reviewing HDCS drawings? - Provide checklist for review of Communication System drawings. Expected Outputs - System-wise Design Review Checklists. - Risk Identification Report. - Design Review Recommendations. - Rule/Document References.	f) Providing intelligent suggestions to designers during design and review activities. g) Continuously improving recommendations through newly added project data and user feedback. Example User Prompts: - Generate Design Review Checklist for MB/SRE System. - What are the common design issues observed in previous projects for HDCS? - What risks should be considered while reviewing Integrated Bridge System drawing? Expected Outputs - Project-wise Design Review Checklists. - System-wise Design Review Checklists.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Excel, Word, and PDF Outputs.	- Discipline-wise Design Review Checklists. - Risk Analysis & Mitigation Report. - Design Review Recommendations - Rule/Document References. - Excel, Word, and PDF Outputs.					
12	Workspace Management	Hierarchical workspace, project-wise repositories, document upload, retrieval and search functionality.	HSL presently maintains all documents on an existing server. A separate AI server with GPU capability shall be provided for hosting the AI application. The system shall be capable of: - Integrating with the existing HSL document repository without duplication of documents. - Accessing, indexing, and retrieving documents stored on the existing server while the AI application resides on a separate AI server.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			c) Providing a logical hierarchical workspace structure for users to access documents project-wise, discipline-wise, system-wise, and document-type-wise. d) Mapping documents available in the existing repository to the required workspace hierarchy using folder structure, metadata, naming conventions, or other suitable mechanisms. e) Allowing users to browse, search, filter, select, and process documents from the workspace based on their access privileges. f) Automatically identifying and indexing newly added documents for inclusion in the AI knowledge base. g) Supporting role-based access control and maintaining audit logs for					

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VENDOR INFORMATION, CLARIFICATION REQUIREMENTS AND MVP COMPLIANCE MATRIX

S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			document access, searches, uploads, downloads, and AI interactions. h) Ensuring synchronization between the existing document repository and the AI knowledge base. HSL Responsibility a) Provide access to the existing document repository and associated permissions. b) Provide the desired workspace hierarchy and document classification structure during implementation. c) Support identification of project-wise, discipline-wise, system-wise, and document-type-wise document categorization wherever required. Vendor Responsibility					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			(a) Configure and implement the workspace hierarchy within the AI application. (b) Map and organize documents from the existing repository into the required logical workspace structure without physically relocating or duplicating documents. (c) Develop the required indexing, search, retrieval, and synchronization mechanisms. (d) Ensure seamless access to documents stored on the existing server from the AI application hosted on the AI server. Vendor Confirmation Required (a) The proposed solution can operate with documents residing on the existing HSL server while the AI application is					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			hosted on a separate AI server. (b) No duplication of the complete document repository is required for normal operation of the system. (c) The RAG framework can continuously ingest and process newly added documents after deployment without requiring major redevelopment. (d) The proposed architecture supports future expansion of the HSL document repository and knowledge base.					
13	Audit Log Management	The AI tool shall demonstrate the capability to maintain audit logs for key user activities within the application.	The system shall maintain comprehensive audit logs for all user activities performed within the application. The system shall be capable of:					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		The system shall be capable of: a) Recording user login and logout activities. b) Recording user searches, AI queries/prompts, and generated responses. c) Recording document uploads and downloads. d) Recording updates made to records such as inspection remark status and feedback submissions. e) Capturing user identity, date, time, module accessed, and action performed.	a) Recording user login, logout, and authentication activities. b) Recording document uploads, downloads, updates and deletions. c) Recording user searches, AI queries, prompts, and generated responses. d) Recording creation, modification, and approval of data, reports and outputs. e) Recording status updates made to inspection remarks and other workflow items. f) Capturing user identity, date, time, module accessed and action performed. g) Providing authorized administrators with facilities to search, filter, view, and export audit logs.					

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		f) Allowing administrators to view and filter audit logs. Expected Outputs - Login/Logout History. - User Search & Query History. - Document Activity Log. - Status Change History. - Audit Log Report.	h) Maintaining audit logs in a secure and tamper-resistant manner. i) Supporting audit log retention as per HSL requirements. Expected Outputs - User Activity Log. - Login/Logout History. - Search and Query History. - Document Access Log. - Data Modification Log. - Status Change History. - Exportable Audit Reports.					
14	User Interaction History	The AI tool shall demonstrate the capability to maintain and retrieve the history of user interactions within the application. The system shall be capable of: a) Recording user searches, queries/prompts,	The system shall maintain a comprehensive history of user interactions, AI-assisted activities and generated outputs across all modules. The system shall be capable of: a) Recording and maintaining user searches, queries/prompts, AI responses, generated					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		and AI-generated responses. b) Maintaining a history of user interactions across different modules. c) Allowing users to view and revisit their previous interactions. d) Providing search and filter options based on date, module and keywords. e) Allowing users to reuse or continue previous interactions. Expected Outputs • User Query History. • Search History. • AI Interaction History.	reports and document processing activities. b) Maintaining project-wise, system-wise, module-wise, and user-wise interaction history. c) Allowing users to search, filter, retrieve, revisit, and continue previous interactions. d) Providing access to previously generated outputs, reports, analyses, and recommendations. e) Supporting role-based access control for viewing and managing interaction history. f) Providing administrators with monitoring, analytics, and reporting capabilities for user interactions and system usage. g) Maintaining complete traceability of AI-generated outputs with					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Filtered Activity Reports. etc	associated source references and timestamps. h) Retaining interaction history as per HSL requirements and supporting export of records in standard formats. Expected Outputs - User Query & Prompt History. - AI Response History. - Generated Report & Output History. - Project-wise and Module-wise Interaction History. - Searchable User Activity Records. - User Activity Analytics & Usage Reports.					
15	Role-Based Access Control (RBAC) &	The AI tool shall demonstrate role-based and department-based access control capabilities.	The system shall provide comprehensive user management and role-based access control to ensure secure access to documents,					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
	User Management	The system shall be capable of: (a) Supporting multiple user roles such as Administrator and User. (b) Authenticating users through secure login mechanisms. (c) Restricting access to modules, documents, projects, and functionalities based on user roles and departments. (d) Supporting department-wise access control (e.g., Electrical, Machinery, Hull, Outfit, Piping, Production, QA,	knowledge repositories, AI functionalities, and administrative features. The system shall be capable of: (a) Supporting configurable user roles, departments and permissions as defined by HSL. (b) Authenticating users through secure login mechanisms. (c) Controlling access to projects, systems, disciplines, departments, documents, reports, dashboards and AI functionalities based on assigned roles and permissions. (d) Supporting department-wise access control and document visibility restrictions. (e) Providing administrators the ability to create, modify, activate, deactivate and manage					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		Planning, Submarine, Ship Repairs etc.). (e) Providing administrators the ability to create, modify, activate, and deactivate user accounts. (f) Ensuring users can access only authorized data, documents and AI functionalities. (g) Maintaining records of user access and activities. Expected Outputs • User Login Management. • Role-Based & Department-Based Access Control.	user accounts, departments, roles, and permissions. (f) Restricting access to sensitive documents and information based on authorization levels. (g) Maintaining complete traceability of user access and activities through audit logs. (h) Supporting role-based and department-based visibility of dashboards, reports and workspace content. (i) Providing reports on users, departments, roles, permissions and access activities. Expected Outputs • User & Role Management Dashboard. • Department-wise Access Control Matrix. • User Permission Matrix. • Access Control Reports.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		- User Account Management. - Access Control Reports.	- User Activity Reports. - Role-wise and Department-wise Access Summary.					
16	User Feedback Mechanism	Feedback option shall be available for every functionality with Satisfied / Partially Satisfied / Not Satisfied and User Comments.	The AI tool shall provide a feedback mechanism across all functionalities to capture user satisfaction and support continuous improvement. The system shall: (a) Allow users to provide feedback for every AI response, report, recommendation and generated output. (b) Provide feedback options such as: ○ Satisfied ○ Partially Satisfied ○ Not Satisfied (c) Allow users to submit comments, suggestions, corrections, and improvement requests.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			(d) Provide an Admin Dashboard to monitor and review all feedback. (e) Generate functionality-wise and project-wise feedback analytics and reports. (f) Enable tracking and closure of feedback by administrators. (g) Maintain audit logs of all feedback submissions and actions taken. (h) Support incorporation of approved feedback into the knowledge base during periodic updates. (i) Export feedback reports in Excel, Word, and PDF formats. The feedback mechanism shall support continuous improvement of AI responses, knowledge repositories and overall system performance.					

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17	Ship Cost Estimation Assistant(included for second phase selection)	The AI tool shall demonstrate the capability to generate a preliminary ship cost estimate using the AI-generated Bill of Materials (BoM) generated from the Build Specification/RFP. The system shall be capable of: a) Utilizing the AI-generated BoM as the input for ship cost estimation. b) Identifying the approved vendors for the equipment/materials from a sample vendor database. c) Automatically preparing	The system shall function as an intelligent Ship Cost Estimation Assistant by utilizing the AI-generated Bill of Materials (BoM), Budgetary Quotations (BQs), SAP price data, historical project data, engineering calculations, and HSL-defined costing methodologies. The system shall be capable of: a) Utilizing the AI-generated BoM as the basis for system-wise, discipline-wise and overall ship cost estimation. b) Automatically identifying approved vendors from the vendor database and generating vendor-wise Budgetary Quotation (BQ) enquiry documents/emails for Administrator review and external dispatch.					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		vendor-wise Budgetary Quotation (BQ) enquiry documents/emails based on the generated BoM. d) Generating enquiry documents/emails in a format suitable for review and dispatch by the Administrator through HSL's external email system. e) Demonstrating preliminary ship cost estimation using representative (dummy) unit rates for the generated BoM.	c) Maintaining a vendor-wise enquiry register indicating enquiry status, reminder status, BQ receipt status, pending quotations, and quotation validity. d) Integrating with HSL's SAP system for retrieval of the latest equipment and material prices, subject to HSL providing the required APIs/interfaces. Vendors shall confirm compliance for SAP integration. e) In the event that direct SAP integration is not feasible or successful, the system shall be capable of utilizing equipment and material price data exported from SAP (e.g., Excel/CSV or other agreed formats) and placed in the AI workspace by HSL,					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		f) Demonstrating automatic updating of the ship cost estimate using sample Budgetary Quotations uploaded into the workspace. g) Providing a system-wise, discipline-wise, and overall ship cost summary with complete traceability of quantities, unit rates, assumptions, and cost calculations.	without requiring any modification to the cost estimation logic. f) Automatically utilizing the latest Budgetary Quotations (BQs) uploaded into the workspace to update equipment and material prices wherever applicable. g) Applying HSL-defined engineering costing logic and man-day estimation methodology (will be provided during development) for estimating different costs. h) Automatically regenerating system-wise, discipline-wise, and overall ship cost estimates whenever the BoM, SAP price data, Budgetary Quotations, or costing parameters are updated.					

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		<i>Demonstration Workflow</i> AI-generated BoM → Identification of Approved Vendors → Automatic Preparation of BQ Enquiry Documents/Emails → Administrator Review & External Dispatch → Upload of Sample BQs into Workspace → AI-based Ship Cost Estimation & Cost Update <i>Expected Outputs</i> • Vendor-wise BQ Enquiry Documents. • Preliminary Ship Cost Estimate. • System-wise & Discipline-wise Cost Break-up. • Updated Cost Estimate using Uploaded BQs.	i) Providing complete traceability for every cost element by indicating the quantity source (BoM), price source (SAP/BQ/Workspace), engineering calculations, assumptions, and costing methodology adopted. j) Maintaining version history of cost estimates and enabling comparison of revisions generated using updated SAP prices or Budgetary Quotations. <i>Vendor Confirmation Required</i> The vendor shall confirm that: • The proposed solution supports integration with HSL's SAP system through APIs/interfaces, if made available by HSL. • If direct SAP integration is not available or is unsuccessful, the solution shall seamlessly utilize SAP price data exported by HSL and placed in the AI workspace, without any					
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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
		- Cost Traceability Report. - Excel, Word, and PDF Outputs.	change in functionality or additional development effort. - The same cost estimation engine shall function irrespective of whether the price data is obtained directly from SAP or through SAP-exported files. - The proposed solution shall support future SAP integration whenever APIs/interfaces become available, without requiring redevelopment of the core application.					
18	Additional		During the development phase, the following capabilities may also be required as part of the AI-Driven Design Validator. Vendors shall confirm that the proposed solution is capable of supporting these functionalities without additional cost implication:					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			(a) HAT/SAT Protocol Generation: Generate Harbour Acceptance Test (HAT) and Sea Acceptance Test (SAT) protocols for systems in the prescribed HSL format using knowledge base and past Protocols. (b) Comprehensive Onboard Spares (OBS) / material List Generation in prescribed format : Automatically generate a comprehensive Onboard Spares (OBS) /material List for a project in prescribed format based on the Bill of Materials (BoM) specified in the Purchase Order Technical Specification (POTS) of all the equipment/systems belong to one project. Note: The above functionalities shall be considered as					

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S.No	Feature	HSL Expectation During MVP	HSL Expectation After Development	Current Product	Customization Required	New Development required	Vendor compliance	Remarks
			enhancements within the approved modules and shall not be construed as separate core modules.					

3. DOCUMENTS TO BE SUBMITTED AS PART OF TECHNICAL OFFER /ADDITIONAL INFORMATION REQUIRED

- a) Technical proposal with Architecture Diagram of Proposed Solution and AI model details
- b) Screenshots of MVP Demonstration with corresponding Sample Output Files generated during MVP (.docx, .xlsx and .pdf) and video of MVP
- c) Work Completion Certificates for Government/PSU/Defence Projects etc
- d) Client References.
- e) Resource Deployment Plan.
- f) Project Development Schedule.
- g) Training and Knowledge Transfer Plan.
- h) Company's Registration Certificate
- i) Technical specification document compliance report (shared by HSL)
- j) Assumptions, Limitations and Dependencies.

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VENDOR DECLARATION

We hereby certify that the information furnished above is true and correct. We understand that the above information may be used by HSL for technical evaluation and assessment of the proposed AI Driven Design Validator solution.

Authorized Signatory:

Name:

Designation:

Company Seal: